Laura Martin: Real Options and the Cable Industry

On May 4, 1999, Laura Martin paced in the Rainbow Room of the Rockefeller Center in New York City as she prepared for her presentation at the Credit Suisse First Boston (CSFB) 1999 Broadband conference. She had assembled a panel of four leading figures from the U.S. cable industry to address a group of 400 institutional money managers on the state of the industry. On her panel were the CFO of Adelphia Communications, the CEO of AT&T's Cable Operations, the Vice Chairman of Comcast Corporation, and the CEO of Cox Communications. As the equity research analyst for cable stocks at CSFB, a global investment banking firm, Martin realized that this meeting afforded her a unique opportunity to demonstrate her knowledge of the drivers of value in the cable industry. The stakes were even higher as she had chosen to unveil a new approach to valuing cable stocks at this panel. This "real options" valuation metric would be new to both the panel and the audience, and Martin wondered if they would appreciate a new way to think about their industry – or if they would simply dismiss it.

The cable industry historically generated revenues exclusively from providing analog video. Rapid advances in digital and compression technologies were now creating new potential sources of revenue including digital video, cable telephony, and high-speed Internet access. Additionally, the industry, once the domain of family-owned local operators, was consolidating quickly and the boundaries between the cable industry and other media industries were quickly dissolving.

Martin had observed these changes as a sell-side analyst for CSFB covering the cable industry for the last five years. During that time, she had tried to differentiate herself from her competitors through an emphasis on more advanced valuation techniques. While most of her competitors were content with metrics such as Earnings Before Interest, Taxes, Depreciation and Amortization (EBITDA) multiples, Martin had chosen to emphasize discounted cash flow analyses and Economic Value Added analyses. Recently, her attention had shifted to real options analysis as she felt other valuation metrics neglected an important aspect of the cable valuation puzzle. She had chosen this conference as the moment to unveil the implications of real options analysis to the companies she evaluated and the money managers she served as clients.

As she prepared to present her findings, she reviewed the details of her analyses and considered how she could best ensure the acceptance of her ideas. Was she correct in thinking of the unused capacity as a real option? Were the inputs to her analysis correct? Most importantly, how could she overcome the resistance to complex valuation metrics in an industry that still emphasized EBITDA multiples–a valuation metric she dismissed as primitive?

Professors Mihir A. Desai and Peter Tufano prepared this case with the assistance of Joshua Musher, Research Associate, as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

Laura Martin and the Equity Research Industry

After graduating from Harvard Business School in 1983, Martin spent seven years as an investment banker at Drexel Burnham Lambert, four years as a "buy-side" analyst at Capital Research and Management, and the last five years as a "sell-side" equity analyst at CSFB.

A "buy-side" analyst typically worked for a money management firm. In her capacity as a buy side analyst, Martin both managed a portfolio of equities and provided her portfolio manager colleagues with stock recommendations for the industries she covered. Research produced by buy-side analysts was solely for the firm that employed the analyst, and was not distributed widely. In contrast, a "sell-side" analyst worked for a broker-dealer that distributed or sold the research to their buy-side clients. CSFB employed 265 equity analysts worldwide who covered eighty-five industries, and another five analysts who covered broad trends in the equity markets.

Martin's primary motivation in moving to the sell-side from the buy-side was to take advantage of the depth of analysis conducted on the sell-side. Martin commented, "On the buy-side I covered five industries and 150 stocks. On the sell-side, I only cover 15 large capitalization stocks and I focus on two industries. In addition, while firms on the buy side typically have a specific investment style, on the sell-side I serve clients with all types of money management disciplines. This requires me to tailor my analysis and recommendations to each client's needs."[1]

Martin considered three constituencies central to her work as a sell-side analyst – the companies she covered, the buy-side firms she advised, and internal CSFB constituencies. Martin estimated that she spent approximately 40% of her time analyzing the firms she covered, 35% communicating that analysis to buy-side clients, and 25% on internal CSFB activities.

- Martin visited each of the companies she covered at least once a year. During those meetings, Martin would typically meet with many company representatives including the CEO, CFO, operating division chiefs (marketing, sales, technology, production, etc) and the head of investor relations. These visits typically lasted one day. In addition, many companies sponsored one-day investor conferences each year that Martin would attend along with 100 other buy-side and sell-side representatives. Also, Martin spoke with company representatives by phone at least every quarter. Finally, Martin interacted with company representatives at industry shows and at conferences.

- Martin regularly communicated via fax, voice mail, or email with approximately 900 individuals employed by the largest institutional money managers in the world, predominantly in the United States. Because CSFB did not have retail operations, Martin did not call on non-institutional clients. Martin believed that she saw or visited approximately 250 of those accounts annually. For the remainder, she would field inquiries over the phone. Martin emphasized that communicating with clients was her mainstay. "I get paid to talk. On the buy-side, you get paid to outperform your index. You don't have to talk to anyone. The sell-side is more about communicating the results of your analysis, either in print or verbally. I publish 1,000 pages each year[2] and I use a

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1 Money managers typically tailored their funds according to particular investment strategies or styles. For example, managers might favor stocks with high expected earnings growth, or "growth" stocks. Alternatively, other managers might favor neglected stocks that were considered relatively cheap, or "value" stocks.

2 Each written report contained a lengthy disclaimer by CSFB, which we reproduce here at their request: "This report is provided to you solely for informational purposes and does not constitute an offer or solicitation of an offer, or any advice or recommendation, to purchase any securities or other financial instruments and may not be construed as such. This report may not be reproduced or redistributed to any other person, in whole or in part, without the prior written consent of the distributor listed below. The information set forth herein has been obtained or derived from sources believed by Credit Suisse First Boston Corporation and its affiliates ("CSFB" or

variety of communication technologies, including blast faxes, blast voicemails and blast emails, to reach my institutional clients." Her reports were typically filled with valuation analyses that supported a target price for a given stock, earnings estimates for that stock, and her investment recommendation (buy, sell or hold).[3]

- Martin considered her relationships with internal CSFB constituencies to be the third important dimension of her job. These internal constituencies were the institutional sales force, the trading desk and the investment banking arm of CSFB. The sales force was an important conduit between Martin and her buy-side clients. When Martin had an important idea to share about any of her companies, she would communicate it on CSFB's 7:30 am daily research call. CSFB's sales force would make thirty to forty calls each to interested clients, rapidly disseminating her message to hundreds of buy-side clients prior to the market opening. Martin also interacted with CSFB's trading desk frequently. CSFB acted as an agent and principal in trading activity; if there was unusual behavior in a one of Martin's stocks, a trader might call Martin to ask her to follow up with the company or discuss rumors that might be affecting the stock. Investment banking was Martin's third internal constituency and an important CSFB profit center. If any of Martin's companies issued equity, Martin was expected to answer questions regarding valuation and the overall business for any clients that were considering participating in the deal. In merger and acquisition activity, Martin was sometimes asked for expert advice about how the market would react to specific deal structures or pricing.

According to Bloomberg, 29 equity research analysts covered the broadcasting/cable industry.[4] Martin sought to differentiate herself from other equity research analysts through the depth of her analyses and the use of novel valuation techniques. Among the analysts who covered the industry, Martin suggested that "everyone carves out different niches. I compete on intellectual leadership and depth of analysis. I have competitors who position themselves as detail folks who publish loads of details while others are relationship types who call every CEO every month." With respect to valuation, virtually all leading analysts reported valuation in terms of EBITDA multiples, with a quarter also reporting multiples of price-to-free-cash flow, and about a quarter reporting discounted cash flow analyses.

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[3] According to a Zacks Investment Research study, 67.5% of all analyst recommendations were buys, 1.4% sells, and 31.1% holds. Jeffrey M. Laderman, "Wall Street's Spin Game," *Business Week*, October 5, 1998.

[4] According to Nelson Information Services, there were 3,724 full-time equity analysts. Laderman, *ibid*.

Martin believed that there were two primary drivers to her compensation: her annual ranking in *Institutional Investor* magazine and revenues linked to corporate finance deals. While there was not an explicit formula linking her compensation to these metrics, Martin felt that compensation was quite sensitive to both measures. Each year, *Institutional Investor* polled investors to create an "All-America Research Team." Voters were asked to rank five attributes of all sell side analysts. In 1999, these attributes were ranked as follows: (1) industry knowledge (8.09 out of 10), (2) written reports (7.26 rating), (3) stock selection (7.20 rating), (4) earnings estimates (6.83 rating), and, (5) special services including company visits and conferences (6.70 rating). Martin concurred with the survey's emphasis on industry knowledge and the downplaying of stock selection and earnings estimates. Martin also added that for high ratings, "accessibility is extremely important. I have to basically be available 24 hours a day." Martin was ranked in both industry categories that she covered.

Valuation in the Cable Industry: Cox Communications

Cable television began in the early 1950s in the United States when entrepreneurs established "community antenna television" (CATV) systems in rural areas where over-the-air broadcast reception was poor. By 1955, there were 400 CATV systems with 150,000 subscribers. Cable entrepreneurs subsequently expanded into suburban systems and urban areas. By 1978, cable wires passed a total of 26.8 million households (representing 34% of total U.S. television households) and there were 14.2 million basic cable subscribers paying an average of $7.13 per month for an average of eight channels. By 1999, according to the MPAA, there were 97 million homes passed by cable wires (representing 97% of households) and 69 million cable subscribers paying an average of $38 per month for 45 channels.[5]

In December 1998, Cox was the fifth largest multiple system operator (MSO) of cable systems in the U.S. with 3.78 million basic cable subscribers in its consolidated cable systems which passed 5.97 million homes. The corresponding 63% penetration rate was the highest penetration rate among the five largest MSOs. Cox's cable systems were tightly clustered with approximately 80% of its total subscribers in its ten largest clusters In addition to its cable assets, Cox held stakes in cable programming services and in technology and telephony companies. In valuing a company such as Cox, Martin separately valued each of the companies held and then aggregated the implied equity values with the value of the cable business. Historical financials for Cox Communications are presented in **Exhibit 1**. **Exhibit 2** provides summary financials for several comparable cable companies

According to Martin, the cable industry was on the cusp of dramatic changes. Historically, cable companies delivered only analog video signals to homes that paid for the service (cable subscribers). Technological change had created three new potential revenue sources, all based on digital technology: digital video, high-speed data, and cable telephone. As a result of these new potential profit centers, Martin forecast three changes in fundamentals that would become key value drivers for the cable business in the future:

1. *Revenue growth would accelerate and diversify*: From 1998 to 2003, Martin projected that revenue would grow at a compound annual growth rate (CAGR) of 14.2%. The mix of revenue would shift from nearly 100% analog video revenue in 1998 to an equal division between analog and digital product revenues over the 5-year period. (See **Exhibit 3**.)

[5] For industry history, see Sally Bedell Smith's In all His Glory: The Life of William S. Paley, New York: Simon & Schuster, 1990.

2. *Capital spending would slow and the nature of investments would change:* While upgrading the existing cable plant and maintenance capital constituted the majority of capital spending in 1998 and was expected to increase for the following two years, further capital spending would substantially decrease. What spending remained would be directed toward the new technologies. Consequently, overall capital spending, excluding acquisitions, would diminish at an average CAGR of -6.5%. Martin believed that the shift from non-revenue linked (plant upgrades) to revenue linked (new technologies) capital expenditures implied falling risk for investors.

3. *The digital revenue streams would yield higher returns on invested capital*: According to Martin, the shift in capital spending toward new technologies would generate considerably higher returns on invested capital (ROIC).[6] Martin estimated that marginal ROICs (excluding the cost of upgrading the cable plant) were 45% to 65%, compared to a 1998E ROIC of approximately 5% for Cox. (See **Exhibit 4**.)

Martin also believed that the industry structure of cable was changing. Historically, the cable industry had been the domain of family-owned local operators. Aggressive consolidation within the cable industry and acquisitions of cable companies by non-cable companies (a trend labeled "convergence") implied that the boundaries between the cable industry and other industries (telephone, internet, etc.) were dissolving.

While there were investment risks to the sector and to Cox–including regulatory changes, the difficulty in creating scale economies, and the super-voting power of the Cox family[7]–Martin believed that the changing industry structure would drive price appreciation in the cable sector and for Cox. Martin believed that, in equity research, "unless you can quantify it, you don't get credit for it." Martin employed several methodologies to translate the new fundamentals and industry structure into valuation metrics for the cable stocks, and estimate the potential price appreciation of Cox.

ROIC Target Price Analysis: Historically, cable valuations had closely tracked ROIC according to Martin. Using regression analysis, Martin analyzed the relationship between ROIC and the valuation of cable and entertainment companies as defined by the ratio of enterprise value to average invested capital.[8] (See **Exhibit 5**.) Martin projected that Cox would improve its ROIC by 0.8%, or 80 basis points, in 1999. Martin then used the regression line to estimate the target enterprise value to average invested capital multiple.[9] Adjusting for non-consolidated assets, other assets, cash and option proceeds, and debt, Martin then inferred a target price for Cox by year-end 1999 of $50. This ROIC "Target Price" Analysis indicated to Martin that Cox had significant upside potential relative to its current stock price of $37.50.

EBITDA Multiples Analysis: Martin also framed her target price in the language of EBITDA multiples, a common metric in the cable business. **Exhibit 6** compares how Cox currently traded as a multiple of EBITDA relative to how it would trade if Martin's target price of $50 and her projections for EBITDA were realized. According to this analysis, Cox currently traded at 13.3X EBITDA and Martin's target price would translate into a 20.9X EBITDA multiple for 1999. While these analyses were a mainstay in the cable industry, Martin didn't think they "asked the right questions. They're

6 Martin calculated ROIC by dividing net operating profit after taxes by the average invested capital for the period. Invested capital is the sum of fixed assets and net working capital.

7 Cox Enterprises, owned by the Cox family, controlled 76% of the vote of Cox, mostly through a class of supervoting stock that had ten votes per share.

8 Regression analysis is performed to determine the statistical relationship between variables of interest.

9 Enterprise value was calculated by summing the market capitalization of the equity with book value of debt. Invested capital was the sum of fixed assets and net working capital.

not predictive; they really just reflect the past. Finally, they don't account adequately for the dynamics of balance sheet items which are vital to a capital intensive industry like cable."

Discounted Cash Flow Analysis: Martin's preferred method of valuing companies was the discounted cash flow (DCF) method employed in **Exhibit 7**. Martin preferred the DCF method as she felt "that it forces me to ask better questions." In particular, Martin felt that the explicit treatment of capital spending and depreciation was important for valuing cable companies and that the DCF analysis forced her to think through the dynamics of EBITDA growth for the company. Martin forecasted ten years of EBITDA growth rates for the company and industry, changes in the competitive and regulatory environments, technological shifts, emerging industry structures and value drivers, and other factors that would affect long-term growth. She typically tried to be conservative in her estimates. Similarly, Martin forecast ten years of capital spending and depreciation to arrive at an unlevered "free cash flow" (or cash flow available for all capital providers) for each year through 2008. Using a terminal multiple of 13 times EBITDA and a weighted average cost of capital (WACC) of 9.3%,[10] the DCF analysis implied that Cox's stock was worth $54.29. Again, this represented substantial appreciation relative to the current stock price.

Other methods: In addition, Martin reported two other valuation measures. In her *private market valuation analysis*, she took into account the takeover or control premium for Cox's holdings of non-public assets. In a takeover, these assets might sell for considerably more than their public market value. The private market analysis also provided valuations of all non-consolidated assets. Based on this analysis, she calculated a theoretical value of Cox of about $51. In her *competitive advantage period analysis*, she calculated how long Cox could sustain returns above its cost of capital. She estimated that Cox could sustain a 15-year period of above-market returns, implying a theoretical value well above the current price of $37.50.

Real Options Valuation Analysis

While Martin felt that discounted cash flow analysis was a significantly better valuation tool than EBITDA multiples, she perceived several shortcomings in the new digital world of the cable industry. In particular, Martin noted that the previous analysis did not include the value from unused bandwidth capacity.

Cable companies were upgrading their cable infrastructure to have 750 megahertz (MHz) of bandwidth capacity by replacing coaxial cable with fiber optic cables.[11] (Coaxial cable remained in the ground for the last mile to the home as coaxial cable had enormous bandwidth over short distances.) Of the 750 MHz of capacity in an upgraded cable plant, 550 MHz was typically devoted to analog video and another 98 MHz was dedicated to digital services, network control, telephony and internet services. The remaining 102 MHz, comprised of seventeen 6 MHz channels, was "dark" or "unlit" and was essentially unused. (See **Exhibit 8.**)

Martin considered the DCF valuation unsatisfactory for these unused channels because "companies are being hit for 100% of the capital spending but we're only counting the visible revenue streams from existing services. That unused capacity could be used for a bunch of new interactive

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10 Martin calculated Cox's cost of equity (10.5%) as the 10-year risk-free rate of 5.12% plus the product of Cox's beta of 1.07 times an estimated risk premium of 4.98%. Martin calculated Cox's after-tax cost of debt as 3.65% by assuming that Cox could borrow at 86 basis points above the risk-free rate and that Cox faced a tax rate of 39%. Finally, Martin employed a debt-to-total capital ratio of 18% in arriving at a WACC of 9.3%.

11 Martin estimated that 66% of Cox's plant would be upgraded to 750MHz and an additional 15% to 550 MHz by June 1999. Martin estimated this to cost $280 per home passed, $40 of which Martin associated with the extra capacity of the stealth tier.

services or services that don't even exist today which we're not giving these companies credit for." For example, technologies under development would allow for video streaming, interactive games, and the implementation of local intranets. Martin called the 102 MHz of unused capacity the "Stealth Tier" as the revenue streams were invisible but, in her estimation, not valueless.[12] Given that Cox's stock price was below the DCF valuation and the DCF valuation did not incorporate the value of the stealth tier, Martin reasoned that the stealth tier was actually being ignored by the market.

Martin turned to real options to value the stealth tier. While familiar with financial options, she had recently read several articles and a book that described the use of financial option-pricing methodologies in broader settings.[13] According to Martin, the stealth tier represented a real option for cable companies such as Cox as they could potentially "light up" the stealth tier as new, currently immature or unknown interactive services were developed. For Martin, the contingent nature of the investment decision and the uncertainty surrounding the ultimate revenue streams made the stealth tier ideal for valuation through real options analysis. Just as financial options give holders the right, but not obligation, to buy securities, this real option gave each upgraded cable company the right, but not the obligation, to obtain revenues from the stealth tier, depending on market conditions. Real options analyses had been used in several industry settings including natural resources extraction and pharmaceutical research. In both cases, managers had rights to pursue investment projects, such as extraction or drug development, without obligations to pursue them.

Financial options were valued using a wide variety of methods, the best known being the Black-Scholes model used to value simple European-style equity options.[14] In order to conduct a real options valuation of the stealth tier, Martin realized that she needed analogous inputs to the inputs used for valuing financial options. Martin considered that the best approach was to value the option of *each* channel of the stealth tier *per* home passed.

- As a proxy for the present value of each channel on the stealth tier, Martin used the average current market value per home passed for a typical cable company of $2,500. She divided $2,500 by the 108 lit channels currently being employed to yield a current value of a channel per home passed of $23.15.

- For the strike price of the option, Martin had to calculate the cost to light up one of the seventeen channels of the stealth tier. The technological costs of actually lighting up the fiber were minimal, probably close to zero. It therefore seemed intuitive to use stealth tier channels for existing applications such as additional analog video channels, and switch over to the new high margin applications when they became available. However, turning off analog channels in the cable industry was typically very costly in terms of customer satisfaction, but this cost was difficult to estimate.[15]

  Consequently, Martin thought she might want to consider the higher opportunity cost of not lighting up the fiber immediately. To estimate these forgone profits, Martin began with the current revenue stream per subscriber of approximately $36 per month or $432

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12 The B-2 Stealth Bomber was deployed by the U.S. Air Force as a long-range bomber that employs a technology allowing it to go undetected by enemy radar.

13 M. Amram and N. Kulatilaka, *Real Options: Managing Strategic Investment in an Uncertain World* (Boston: Harvard Business School Press, 1999)

14 F. Black and M. Scholes, "The Pricing of Options and Corporate Liabilities," *Journal. of Political Economy*; 81 (May-June 1973): 637-659. While European-style options are exercisable only at maturity, American-style options are exercisable anytime prior to maturity.

15 For example, one industry executive on Martin's panel told the story of how his company had used an extra channel to show live programming using a camera pointed at a garbage dumpster. When the cable company replaced this with a legitimate cable channel, the executive complained of receiving irate mail from viewers and bad press for canceling "the garbage channel."

per year. A 61% penetration rate implied $263 of annual revenue, or $2.44 ($263/108 channels) per channel. Using a typical cable industry margin of 50%, this translated into $1.22 of forgone annual profit per channel per home passed.

- Martin derived an implied volatility from a traded at-the-money call option on the Cox stock to estimate the relevant volatility for the real option. A one-month call option traded in the market was priced to imply a volatility 50% per year.

- Martin used the life of the cable plant, widely regarded to be 10 years, as the term to maturity.

- For an interest rate, Martin collected information on U.S. Treasury securities. (See **Exhibit 9.**)

These inputs were employed by Martin to estimate an option value of $22.45. (See **Exhibit 10.**) Martin multiplied this result by the number of channels in the stealth tier to arrive at a total value of $381.65. Martin acknowledged, "the seventeen channels should be worth more than just the sum, but I'd prefer to understate the value just to make sure that clients accept the number. It's important to be conservative so that clients don't get lost quibbling with assumptions since my primary goal is to get my clients to consider whether DCF is enough." After netting out the $40 of incremental capital costs Martin associated with installing the stealth tier, corresponding to the premium that the cable company paid for the option, Martin arrived at a net value of the stealth tier per home passed of $341.65. This figure represented an additional 14% premium to the current market valuation per home passed.

Conclusion

While concerned with the individual inputs, Martin was more concerned about the overall message for her clients. Martin noted, "80% of my clients have never heard of this stuff and are still using EBITDA multiples. What I would like to convey is that current valuation methodologies, even DCF, don't give any value to the stealth tier." More generally, Martin was convinced of the value in using real options analysis: "With the New Economy, real options go much further in capturing the nature of investment opportunities. I really believe that real options analysis makes you view companies differently and, as a result, you ask better questions about a company's strategic position and long-term value drivers. For me, the primary goal of all the valuation work that I do is about asking the right question." Having concluded her presentation, Martin turned to the task of selling her analysis, and its implications, to her client base.

Exhibit 1 Cox Communications Historical Financials, 1993-1998

Selected Balance Sheet Items ($Mil.)	1993	1994	1995	1996	1997	1998
Gross Property, Plant and Equipment	$ 1,104.5	$ 1,234.9	$ 2,081.5	$ 2,343.0	$ 2,840.9	$ 3,702.3
Accumulated Depreciation	516.0	570.7	867.6	811.2	861.9	1,050.1
Net Property, Plant, and Equipment	588.5	664.3	1,213.9	1,531.8	1,979.1	2,652.2
Investments at Equity	265.8	534.6	647.2	713.5	795.2	90.7
Other Investments	27.1	34.4	554.0	505.6	854.0	5,890.4
Intangibles	565.9	542.7	2,775.9	2,729.0	2,458.7	3,959.9
Other Assets	34.6	53.5	207.2	139.8	93.2	88.3
Total Assets	1,527.4	1,874.7	5,555.3	5,784.6	6,556.6	12,878.1
Long Term Debt	595.6	787.8	2,575.3	2,881.0	3,141.2	3,885.3
Deferred Taxes	100.1	97.1	288.0	294.5	721.6	2,886.6
Other Liabilities	37.2	56.1	147.7	117.0	110.9	227.2
Preferred Stock	0.0	0.0	0.0	0.0	0.0	2.4
Common Stock	0.0	0.0	270.2	270.3	271.1	277.4
Capital Surplus	581.0	670.5	1,739.4	1,742.1	1,790.8	2,152.3
Retained Earnings	135.6	164.3	322.4	248.9	295.4	2,944.5
Total Equity	716.6	834.8	2,332.0	2,261.3	2,357.3	5,376.6
TOTAL LIABILITIES AND EQUITY	1,527.4	1,874.7	5,555.3	5,784.6	6,556.6	12,878.1

Income Statement ($ Mil.)	1993	1994	1995	1996	1997	1998
Sales	$ 708.0	$ 736.3	$ 1,286.2	$ 1,460.3	$ 1,610.4	$ 1,716.8
Operating Income Before Depreciation	295.6	268.5	498.4	556.9	609.8	659.1
Depreciation, Depletion & Amortization	116.0	128.8	267.3	335.2	404.5	457.7
Operating Profit	179.7	139.8	231.1	221.7	205.3	201.4
Interest Expense	12.9	46.1	146.5	189.3	202.1	223.3
Non-Operating Income/Expense	-23.7	-41.3	-64.6	-115.7	-400.5	-534.1
Special Items	0.0	0.0	183.7	54.7	207.4	2,649.5
Pretax Income	143.1	52.3	203.7	-28.5	-190.0	2,093.5
Total Income Taxes	66.0	25.8	99.9	23.0	-53.5	822.8
Income Before Extraordinary Items & Discontinued Operations	77.1	26.6	103.8	-51.6	-136.5	1,270.7
Extraordinary Items	20.7	0.0	0.0	0.0	0.0	0.0
Discontinued Operations	0.0	0.0	0.0	0.0	0.0	0.0
Net Income	97.8	26.6	103.8	-51.6	-136.5	1,270.7

Source: Cox Communications Annual Reports

Exhibit 2 Summary Financials for Selected Comparable Companies, December 31, 1998 ($ millions)

	Adelphia[a]	Comcast	Cox	Time Warner[b]	Media One
Current Price (4/26/99)	$67.88	$66.00	$37.50	$74.82	$78.82
Fully Diluted Shares Outstanding (millions)	131.4	400.0	565.2	1379.0	663.8
Market Capitalization (4/26/99)	$8,919	$26,400	$21,195	$103,177	$52,321
Total Debt	$3,527	$5,578	$3,920	$10,944	$5,422
Preferred Stock	$648	$0	$0	$575	$1,100
Cash & Option Proceeds	($229)	($982)	($31)	($640)	($915)
Adjusted Enterprise Value	$16,903	$30,882	$25,254	$120,615	$58,457
Non-Cable Consolidated Assets	($999)	($5,914)	($400)	($60,000)	$0
Non-Consolidated Assets	($697)	($7,419)	($11,600)	($20,000)	($31,950)
Implied Cable Value	$15,207	$17,549	$13,254	$40,615	$26,508
Homes Passed (millions)	7.5	7.4	5.9	17.5	7.8
Current Market Value Per Home Passed	$2,041	$2,371	$2,246	$2,320	$3,400
Implied Cable Value + Non-Cable Consolidated Assets	$16,206	$23,463	$13,654	$100,615	$26,508
Adjusted 1999E EBITDA	$1,293	$1,583	$800	$4,816	$1,260
Implied Cable Value + Non-Cable Consolidated Assets /1999E Adj. EBITDA	12.5	14.8	17.1	20.9	21.1
Other Financial Data					
Equity Beta	0.75	1.08	1.07	0.88	1.08
Historic Equity Volatility	57%	51%	47%	44%	NA
Implied Volatility from Options	66%	59%	50%	41%	NA

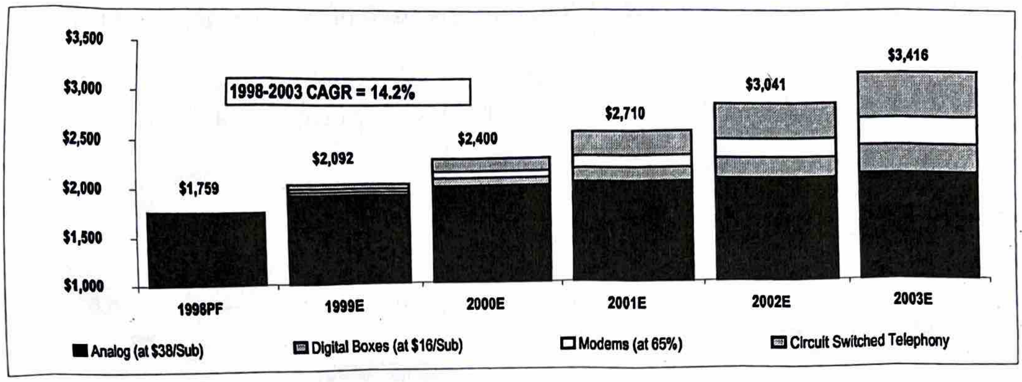
Source: Bloomberg, Credit Suisse First Boston and casewriter estimates. The information is being provided for educational purposes only, and CSFB has not undertaken any review of the suitability of the information or any recommendations contained therein for the recipients of this publication.

[a] Pro forma estimates include announced FrontierVision Partners, Century Communications, and Harron Communications acquisitions, which were expected to close in the second half of 1999.

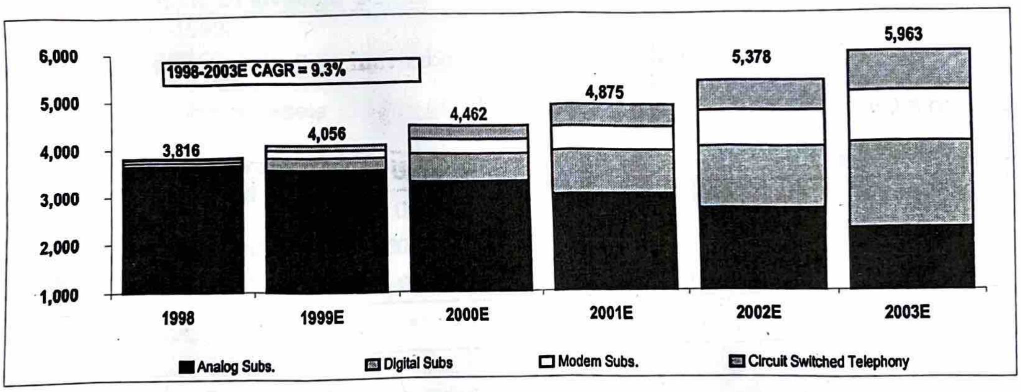
[b] Includes 100% of Time Warner Enterprises.

Exhibit 3 Forecasted Revenue and Subscriber Growth for Cox Communications, 1998-2003

Cable Revenue Projections 1998-2003E



Subscriber Projections, 1998-2003E



Source: Credit Suisse First Boston estimates, published in Laura Martin, Cox Communications (Oct. 12, 1999 Credit Suisse First Boston Corporation.) The information is being provided for educational purposes only, and CSFB has not undertaken any review of the suitability of the information or any recommendations contained therein for the recipients of this publication.

Exhibit 4 Forecast ROICs for New Investments by Cox Communications

Marginal returns on invested capital for new products such as compression boxes, modems, and wireline telephone appear to be significantly higher than the cost of capital. However, *total* returns on capital (including infrastructure upgrade) will depend on the level of consumer demand, consumer price sensitivity, and new product profit margins, each of which is unknown today.

Digital compression boxes Returns are enhanced because digital box costs plus an 11.25% return are legislatively allowed to be recovered over the entire subscriber base over a five-year period.

Marginal ROIC on a Digital Compression Box

Average Cost of Compression Box to Cable Operator	$350
Annual Income:	
Recovery of Box Cost ($6.50/month x 100% profit)	$ 78
Incremental Revenue ($14/month x 50% profit)	$ 84
Annual Income	$162
ROIC/Box	46%

Cable modems Assumes a monthly fee to cable modem subscribers of $40 at a 50% margin.

Marginal ROIC on a Cable Modem

Cost of Modem to Cable Operator	$400
Annual Revenue/Subscriber ($40/month)	$480
Assumed Profit Margin	50%
Profit Per Modem	$240
ROIC/Modem	60%

Wireline telephone Based on Cox Communications' experience in Orange County, California, with a typical monthly bill for telephone customers of $55 at a 56% profit margin.

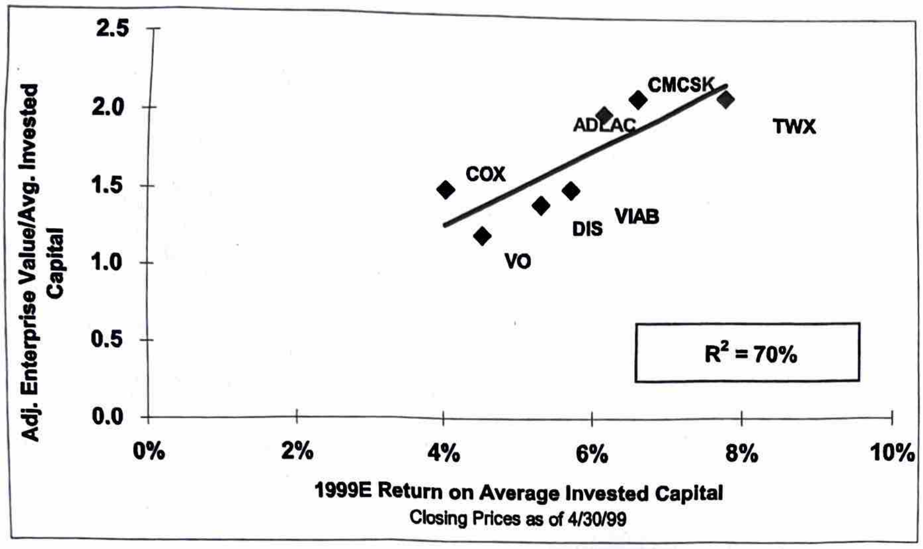
Marginal ROIC on Wireline Telephone

Marginal Cost to Cable Operator	$825
Average Revenue Per Customer[a]	$660
Direct Costs[b]	132
Employees	60
Customer Service	12
Marketing	24
Other	60
Profit Per Telephone Customer Per Year	372
ROIC/Telephone	45%

Source: Credit Suisse First Boston estimates, published in Laura Martin, *Cox Communications* (Oct. 12, 1999 Credit Suisse First Boston Corporation.) The information is being provided for educational purposes only, and CSFB has not undertaken any review of the suitability of the information or any recommendations contained therein for the recipients of this publication.

[a] Includes primary lines, 30% additional lines, access 22, voice mail ports, interconnect trunks, 411, etc.
[b] Includes billing, property taxes, bad debts, powering, etc.

Exhibit 5 ROIC Target Price Analysis for Cox Communication



Based on Multiple Of Invested Capital	1999E
Target multiple-1999E	1.594
Average Invested capital	$12,136
Implied enterprise value	$19,345
Plus: Nonconsolidated assets	$12,292
Plus: Other assets	$400
Plus: Cash & option proceeds at year end	$23
Less: Debt at year end	$(3,800)
Implied equity value	$28,260
Shares outstanding (millions)	565.2
Target price	$50.00
Current price	$37.50
(Target-current)/target	25%
(Target-current)/current	33%

Source: Credit Suisse First Boston estimates, published in Laura Martin, Cox Communications (Oct. 12, 1999 Credit Suisse First Boston Corporation.) The information is being provided for educational purposes only, and CSFB has not undertaken any review of the suitability of the information or any recommendations contained therein for the recipients of this publication.

Exhibit 6 EBITDA Multiple Analysis for Cox Communications, 1999-2000

Based on Enterprise Value EBITDA Year Ended 12/31:	1999E	2000E	Target Price of 2000E EBITDA	Discount from Target Price	Price Appreciation to Target Price
Current share price (4/26/99)	$37.50	$37.50	$50.00	25%	33%
Shares outstanding at year end	565.2	565.2	565.2		
Market capitalization	$21,195	$21,195	$ 28,260		
Plus: Debt at year end	$ 4,090	$ 3,800	$ 3,800		
Less: Non-consolidated Assets	(11,600)	(12,292)	(12,292)		
Less: Other assets (see PMV)	(400)	(400)	(400)		
Less: Cash at year end	(31)	(23)	(23)		
Implied Cable Value	13,254	12,280	19,345		
EBITDA	$ 800	$ 924	$ 924		
Implied Cable Value/EBITDA	16.6	13.3	20.9		

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Exhibit 7 Discounted Cash Flow Analysis for Cox Communications ($ and shares in millions, except per share data)

	1998	1999E	2000E	2001E	2002E	2003E	2004E	2005E	2006E	2007E	2008E	CAGR 99-08
Total EBITDA	$659	$800	$924	$1,047	$1,214	$1,408	$1,634	$1,895	$2,198	$2,550	$2,958	16%
Less: Depreciation & Amortization	($458)	($607)	($680)	($674)	($644)	($616)	($591)	($569)	($549)	($531)	($515)	-2%
Earnings Before Interest and Taxes	$201	$193	$244	$372	$570	$792	$1,042	$1,326	$1,649	$2,019	$2,443	33%
Less: Taxes (EBIT + Amortization at 35%)	($100)	($104)	($123)	($168)	($237)	($315)	($402)	($502)	($615)	($744)	($893)	27%
Plus: Depreciation & Amortization	$458	$607	$680	$674	$644	$616	$591	$569	$549	$531	$515	-2%
Less: Capital Spending & Investments	($978)	($1,040)	($750)	($587)	($517)	($261)	($261)	($261)	($261)	($261)	($261)	-14%
Unlevered Free Cash Flow	($419)	($344)	$51	$292	$460	$833	$970	$1,133	$1,323	$1,545	$1,804	
Terminal Value (TV) of 2008E Total EBITDA											$38,454	

Equity Valuation Analysis	Value	% DCF
PV of Unlevered Cash Flow	$4,456	15%
PV of Terminal Value Discounted to 1999 @ WACC	$17,316	56%
- Long-term Debt at 12/31/99E	($3,800)	-12%
+ Cash at 12/31/99E	$23	0%
+ Non-Consolidated Assets, Private Market Value	$12,292	40%
+ Other Assets, Private Market Value[a]	$400	1%
Discounted Cash Flow Value (DCF)	$30,687	100%

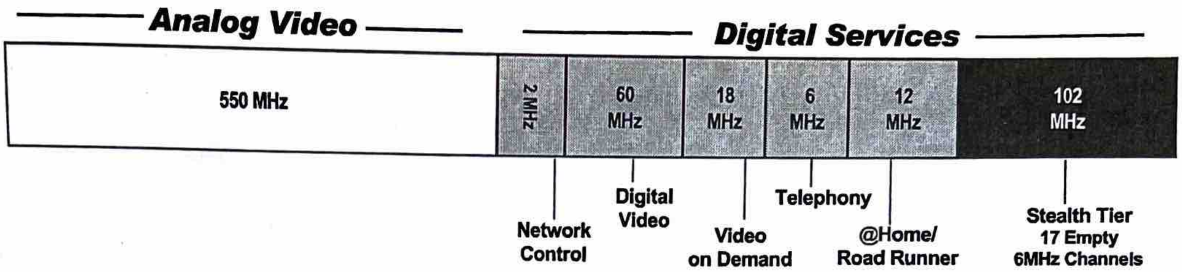
Shares Outstanding @ 9/30/99	565.2
Discounted Cash Flow Value/Share	**$54.29**
Current Share Price (4/26/99)	$37.50
Discount to DCF Value = (DCF-Current Price) / DCF	31%
Price Appreciation to DCF = (DCF-Current Price) / Current Price	45%

Assumptions:	
Discount Rate, Weighted Average Cost of Capital	9.3%
Terminal Multiple	13.0

Source: Credit Suisse First Boston estimates, published in Laura Martin, *Cox Communications* (Oct. 12, 1999, Credit Suisse First Boston Corporation.) The information is being provided for educational purposes only, and CSFB has not undertaken any review of the suitability of the information or any recommendations contained therein for the recipients of this publication.

[a]The Private Market Value took into account the takeover or control premium for Cox's holdings of non-public assets. In a takeover, these assets might sell for considerably more than their public market value.

Exhibit 8 "Stealth Tier" Analysis of 750 MHz Cable Plant Usage



Source: Credit Suisse First Boston estimates, published in Laura Martin, *Cox Communications* (Oct. 12, 1999 Credit Suisse First Boston Corporation.) The information is being provided for educational purposes only, and CSFB has not undertaken any review of the suitability of the information or any recommendations contained therein for the recipients of this publication.

Exhibit 9 U. S. Treasury Interest Rates as of May 1, 1999

Term	Rate
3-Month	4.560%
6-Month	4.657%
1 Year	4.779%
2 Year	5.066%
5 Year	5.219%
10 Year	5.359%

Source: Bloomberg. All yields are quoted on a bond-equivalent-yield basis.

Exhibit 10 Black-Scholes Valuation for Stealth Tier

PV of Potential Project/Home Passed (S)	$23.15
Investment Cost (X)	$1.22
Time to Maturity (T, years)	10
Risk Free Rate (r_f)—	5.25%
Volatility (σ)	50%
Call Option Value/Home Passed	**$22.45**
Times: Number of empty 6 MHz channels on the Stealth Tier	17
Asset Value of the Stealth Tier	$381.65
Less: current investment/home passed	$40.00
Value of the Stealth Tier/home passed	$341.65
Current Market valuation/home passed	$2,500
Stealth Tier/current valuation	14%

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GREGOR ANDRADE

Sampa Video, Inc.

Sampa Video, Inc. was the second largest chain of videocassette rental stores in the greater Boston area, operating 30 wholly owned outlets. Begun in 1988 as a small store in Harvard Square catering mostly to students, the company grew rapidly, primarily due to its reputation for customer service and an extensive selection of foreign and independent movies. These differentiating factors allowed the Sampa Video to compete directly with the leader in the industry, Blockbuster Video. But unlike the larger rival, Sampa had no ambitions to grow outside of its Boston territory. **Exhibit 1** contains summary financial information on the company as of their latest fiscal year-end.

In March 2001, Sampa Video was considering entering the business of home delivery of movie rentals. The company would set up a web page where customers could choose movies based on available in-store inventory and pick a time for delivery. This would put Sampa in competition with new internet-based competitors, such as Netflix.com that rented DVDs through the mail and Kramer.com and Cityretrieve.com that hand delivered DVDs and videocassettes.

While it was expected that the project would cannibalize the existing operations to some extent, management believed that incremental sales would be substantial in the long run. The project would provide customers the same convenience as internet-based DVD rentals for the wider selection of movies available on videocassettes. Sampa also planned to hand deliver DVDs. The company expected that the project would increase its annual revenue growth rate from 5% to 10% a year over the following 5 years. After that, as the home delivery business matured, the free cash flow would grow at the same 5% long-term rate as the videocassette rental industry as a whole. **Exhibit 2** contains management's projections for the expected incremental revenues and cash flows achievable from the project.

Sampa management's major concern was the significant up-front investment required to start the project. This consisted primarily of setting up a network of delivery vehicles and staff, developing the website, and some initial advertising and promotional efforts to make existing customers aware of the new service. Management estimated these costs at $1.5 million, all of which would be incurred in December 2001, as the service would be launched in January 2002.[1]

Management was debating how to assess the project's debt capacity and the impact of any financing decisions on value. In thinking about how much debt to raise for the project, two options were being considered. The first was to fund a fixed amount of debt, which would either be kept in perpetuity or paid down gradually. The second alternative was to adjust the amount of debt so as to

[1] For the purposes of this exercise, it is assumed that all start-up costs would have been capitalized, and depreciated over time. In reality, some of these costs would have been capitalized (e.g., investment in delivery vehicles) while others would have been expensed immediately (e.g., advertising costs).

Professor Gregor Andrade prepared this case. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

maintain a constant ratio of debt to firm value. **Exhibit 3** contains information on market conditions as well as management's assumptions regarding the project's expected cost of debt.

Exhibit 1 Summary Financial Information on Sampa Video, Inc., 2000 (in thousands of dollars)

	FY 2000
Sales	22,500
EBITDA[a]	2,500
Depreciation	1,100
Operating Profit	1,400
Net Income	660

Source: Casewriter estimates.

[a]EBITDA is the Earnings Before Interest, Taxes, Depreciation and Amortization.

Exhibit 2 Projections of Incremental Expected Sales and Cash Flows for Home Delivery Project 2002-2006 (in thousands of dollars).

	2002E	2003E	2004E	2005E	2006E
Sales	1,200	2,400	3,900	5,600	7,500
EBITD[a]	180	360	585	840	1,125
Depreciation	(200)	(225)	(250)	(275)	(300)
EBIT	(20)	135	335	565	825
Tax Expense	8	(54)	(134)	(226)	(330)
EBIAT [a]	(12)	81	201	339	495
CAPX[b]	300	300	300	300	300
Investment in Working Capital	0	0	0	0	0

Source: Casewriter estimates.

[a]EBITD is the Earnings Before Interest, Taxes and Depreciation. EBIAT is the Earnings Before Interest and After Taxes. Taxes calculated assuming no interest expense.

[b]Annual capital expenditures of $300,000 were in addition to the initial $1.5 million outlay, and are assumed to remain constant in perpetuity.

Exhibit 3 Additonal Assumptions.

Risk-free Rate (R_f)	5.0%
Project Cost of Debt (R_d)	6.8%
Market Risk Premium	7.2%
Marginal Corporate Tax Rate	40%
Project Debt Beta (β_d)	0.25
Asset Beta for Kramer.com and Cityretrieve.com	1.50

Source: Casewriter estimates.

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WILLIAM A. SAHLMAN
EVAN RICHARDSON

The Venture Capital Valuation Problem Set

Question 1

Part 1:

John Thompson, CEO of NewVenture, Inc., seeks to raise $5 million in a private placement of equity in his early stage venture. Thompson conservatively projects net income of $5 million in year five and knows that comparable companies trade at a price earnings ratio of 20X.

a. What share of the company will a venture capitalist require today if her required rate of return is 50% *per annum*?

b. If the company has 1,000,000 shares outstanding before the private placement, how many shares should the venture capitalist purchase? What price per share should she agree to pay if her required rate of return is 50%? (Note: Assume investment is in standard preferred stock with no dividends and a conversion rate to common of 1:1)

c. John feels that he may need as much as $12 million in total outside financing to launch his new product. If he seeks to raise the full amount in this round, how much of his company will he have to give up? What price per share will the venture capitalist agree to pay if her required rate of return is 50%?

d. Draw graphs that show the relationship between the investor share required, the investor discount rate and the amount of capital raised.

e. Samantha Jones of Gorsuch Capital likes Thompson's plan, but thinks it naive in one respect: to recruit a senior management team, she believes Thompson will have to grant generous stock options in addition to the salaries projected in his business plan. From past experience, she thinks management should have the ability to own at least a 15% share of the company by the end of year 5. Given her beliefs, what share of the company should Samantha insist on today if her required rate of return is 50%?

Professor William A. Sahlman and Research Associate Evan Richardson prepared this exercise as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

Part 2:

During negotiations between Samantha Jones and John Thompson, Jones proposes using a hybrid security called participating preferred stock. A participating preferred security enables the holder to convert the security to common stock at the previously agreed upon conversion rate **and** to have the principle amount repaid. In standard convertible preferred stock, the holder gets the maximum of the liquidation value **or** the value if converted into common stock.

a. What share of the company will Samantha Jones require today if her required rate of return is 50% and she uses a participating preferred stock instead of a standard convertible preferred security?

b. If the company has 1,000,000 shares outstanding before the private placement, how many shares should Samantha purchase? What price per share should she agree to pay if her required rate of return is 50%?

c. How does utilizing participating versus standard preferred change John and Samantha's perceptions of the risk and reward profile of this deal?

Part 3:

On further analysis and discussion, Samantha and John agree that the company will probably need another round of financing in addition to the current $5 million. Samantha believes that NewVenture will need an additional $3 million in equity at the beginning of year 3. While the first round investors (including herself) will require a 50% return, Samantha feels that round 2 investors, in recognition of the progress made between now and then, will probably have a hurdle rate of only 30%. As before, management should have the ability to own a 15% share of the company by the end of year 5.

a. Based on this new information, what share of the company should Samantha seek today? What price per share should she be willing to pay?

b. What share of the company will the Round 2 investors seek? What price per share will they be willing to pay?

c. How would your answers change if you assume that the 15% for management is allocated at the beginning of the period and that management gets diluted as new shares are issued?

Question 2

Assume a company had an A-round in which it sold 10 million shares of convertible preferred stock at $1.00 per share. Prior to the financing, Founders and Management owned 15 million shares of common stock. The company needs to raise an additional $5 million in a Series B round and learns that the investors insist on a pre-money valuation of $15 million. Your assignment is to figure out the new price per share of Series B convertible preferred stock under three scenarios:

1. There is no dilution protection for prior investors (i.e., there is no price adjustment for prior investors).

2. There is weighted average dilution protection (i.e., old investors receive additional shares – free – to lower their effective price paid per share).

3. There is full ratchet protection (i.e., old investors are issued new shares for free such that their effective price equals the new price being paid by new investors).

Under each scenario, also determine what share of the common stock equivalents outstanding is owned after Series B by Founders and Management, the Series A, and the Series B investors.

Note: a typical formula for calculating the price adjustments using weighted average dilution protection is:

NCP= [(OB*OCP) + New$] / OA

where:

NCP = New Conversion Price

OB = Outstanding Shares Before Offering

OCP = Old Conversion Price

New$ = Amount Raised in Offering

OA = Outstanding Shares After Offering

Question 3

An entrepreneur and a venture capitalist are negotiating a first round of financing and reach an impasse. The entrepreneur believes his company is worth $5 million (equivalent to $1.00 per share given the 5 million fully diluted common shares outstanding before the new round). The venture capitalist is eager to do the deal but believes the correct valuation is only $4 million ($0.80 per share). She wants to invest $4 million at that price and has proposed a simple convertible preferred with weighted average anti-dilution protection. To break the impasse, the venture capitalist proposes to pay $1.00 per share of standard convertible preferred but asks for a 5-year option to invest $3 million in a new convertible preferred stock at a 20% premium (i.e., $1.20 per share).

What valuation is the venture capitalist assigning to the company? What is the equivalent current price per share? If you think of the venture capitalist's option to buy shares in terms of their warrant equivalent, what is the value of those warrants?

Bonus

Dale DePriest looked from one term sheet to the other. It was May 2003, and DePriest had just received two term sheets from Wally Smith of Dillingham Venture Partners (DVP). Smith had agreed to be the lead investor in DePriest's venture, Togiak Networks, and in fact had agreed to seed the writing of the business plan.

The plan was to have one other investor in addition to DVP. DePriest and Smith had agreed that the basic terms should be settled before considering other investment firms. Smith had early on proposed that DVP would be willing to take less equity if DePriest would accept a set of terms generally more favorable to the investors. In one option (A), DVP would invest $3 MM in a simple convertible preferred, with the founders owning 22.5% of the fully-diluted shares that would be outstanding (with a 15% allocation for employee options). In the other option (B), the same amount would be invested in participating preferred. In this option, management would start out owning a little over 29% of the post-financing fully-diluted shares outstanding (with a 19.4% allocation for employee options). DePriest agreed to consider the proposal. The two term sheets are summarized below:

	Option A	Option B
Amount	$3 million	$3 million
Security	Convertible Preferred	Participating Preferred
Mandatory Conversion	Mandatory on IPO > $10 MM, Price>$4.00	Not applicable
Price	$1.00 per share	Same
Liquidation Rights	Liquidation preference on merger, sale or liquidation	Liquidation preference on merger, sale or liquidation
Option Pool	720,000	1,128,000
Option Pool Vesting	5 year straight line with 1 year blackout	Same
Founder Shares	1,080,000	1,692,000
Founder Vesting	20% vesting at close, 5% per quarter	Same

Questions:

1. How would you compare and contrast these two offers as the founders? As employees? As investors?

2. If the company goes public with a terminal market value of $200 MM, how will the pie be split, assuming no more financings? How would the pie be split if the harvest were a merger or acquisition? How sensitive is the split to the actual terminal value?

3. How should the founders, employees, and investors think about the potential implications of future financing rounds?

4. What can you infer about Wally Smith and the other founders from the choice they make? What can you infer about Dale DePriest?

It will be extremely helpful if you spend some time creating an Excel model of this decision. Please include graphs that show how the payoffs to various parties change with the total terminal value of the company.